

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation(BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	20% of CIA	Total Marks:	50 (5 Questions × 10 Marks)
CO2 Statement:	Apply image enhancement and segmentation techniques for extracting meaningful regions from images..(BL3)		
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Section / Batch:	Slot - C	Date:	

Code of Conduct
 I, Y. Navya (192425177) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Navya

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Grey Level Transformations	Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques	Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement	Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement	Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation	Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10
GRAND TOTAL:			50 Marks

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Grey Level Transformation

A grayscale image is enhanced using a negative transformation.

- (a) Explain the purpose of grey level transformation in image enhancement
- (b) Apply the negative transformation:

$$s = 255 - r$$

to the following pixel values:

$$r = 50, 100, 150, 200$$

Compute the transformed pixel values and interpret the effect on the image.

2. Histogram Equalization

A 4-bit grayscale image has the following histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

- (a) Explain why histogram equalization is used for image enhancement.
- (b) Compute the cumulative distribution function (CDF).
- (c) Determine the equalized intensity values and interpret the improvement in image contrast.

3. Gaussian Smoothing

A Gaussian filter is applied to reduce image noise.

- (a) Explain why Gaussian filtering is preferred over averaging filtering.
- (b) Apply the Gaussian kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

to the image region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Compute the new center pixel value and interpret the smoothing effect.

4. Edge Detection Using Sobel Operator

An image is processed using the horizontal Sobel operator.

- (a) Explain the concept and purpose of edge detection.
- (b) Apply the horizontal Sobel mask

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

to the image

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Compute the gradient at the center and interpret the result.

5. Contrast Stretching

A grayscale image has intensity values in the range 50–150.

- (a) Explain the need for contrast stretching in image enhancement.
- (b) Using linear contrast stretching to map the range [50,150] to [0,255], compute the transformed values for

$$r = 50, 100, 150$$

Show the calculations and interpret the improvement in image contrast.

Rubrics for Calculation

	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks	
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem		
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method		
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process		
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations		
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Grey level Transformation

a) Purpose of Grey level Transformation

Grey level transformation modifies the intensity value of pixels to improve image quality.

Purpose:

- Enhance image contrast
- Highlight important features.
- Improve visibility of dark or bright regions.
- Prepare images for further processing.

A negative transformation converts bright areas into dark areas and viceversa.

Formula: $S = 255 - r$

where

r = Original pixel value

S = Transformed pixel value.

b) Apply Negative Transformation

Given,

$$r = 50, 100, 150, 200$$

Using

$$S = 255 - r$$

Original Pixel (r)	Calculation	Transformed Pixel (s)
50	255 - 50	205
100	255 - 100	155
150	255 - 150	105
200	255 - 200	55

Final Answer:

205, 155, 105, 55

2. Histogram Equalization

Given histogram

Intensity	Frequency
0	4
1	6
2	10
3	6

Total pixels

$$N = 4 + 6 + 10 + 6 = 26$$

a) Why Histogram Equalization

- Improves image contrast
- Spreads pixel intensities normally
- Enhances details in low-contrast images.
- Makes dark and bright regions easier to distinguish.

b) Compute the CDF

Step-1: Probability

PDF

Intensity	Frequency	Probability
0	4	$4/26 = 0.153$
1	6	$6/26 = 0.230$
2	10	$10/26 = 0.384$
3	6	$6/26 = 0.230$

Step-2: Cumulative Distribution Function

Intensity	Probability	CDF
0	0.153	0.1538
1	0.230	0.3846
2	0.384	0.7692
3	0.230	1.0000

c) Equalized Intensity values

Since this is a 4-level image

$$L = 4$$

Formula

$$S = (L-1) \times \text{CDF}$$

$$S = 3 \times \text{CDF}$$

Intensity	CDF	Calculation	New Intensity
0	0.1538	$3 \times 0.1538 = 0.46$	0
1	0.3846	$3 \times 0.3846 = 1.15$	1
2	0.7692	$3 \times 0.7692 = 2.31$	2
3	1.0000	$3 \times 1 = 3$	3

Final Equalized Values

Original	Equalized
0	0
1	1
2	2
3	3

3) Gaussian Smoothing

a) Why Gaussian Filtering

- * Reduces random noise effectively
- * Gives more importance to nearby pixels.
- * Produces smoother results.
- * Preserves edges better than simple averaging
- * Avoids blocky artifacts.

b) Apply Gaussian kernel

Kernel $\frac{1}{16} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix}$

Image

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Multiply Corresponding values:

$$= (10 \times 1) + (20 \times 2) + (10 \times 1) + (20 \times 2) + (40 \times 4) + (20 \times 2) \\ + (10 \times 1) + (20 \times 2) + (10 \times 1)$$

$$= 10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10$$

$$= 360$$

Divide by 16

$$\frac{360}{16} = 22.5$$

4) Edge Detection Using Sobel Operator

a) Purpose of Edge Detection

It is used for:

- Object detection
- Image Segmentation
- Feature Extraction
- Boundary Identification

b) Apply Horizontal Sobel Mask

Mask

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Multiply

Top row:-

$$(-1 \times 10) + (-2 \times 10) + (-1 \times 10) = -40$$

Middle row:-

Bottom row:-

$$(1 \times 30) + (2 \times 30) + (1 \times 30) = 120$$

Total:

$$120 - 40 = 80$$

5) Contrast Stretching

a) Need for Contrast Stretching

Benefits:

- Improves visibility

- Enhances Contrast
- Makes details more distinguishable
- Utilises the Complete grayscale range.

b) Linear Contrast Stretching

Given,

Input range : 50 → 150

Output range : 0 → 255

$$\begin{aligned} \text{Formula: } S &= \frac{(x-50)(255-0)}{150-50} \\ &= \frac{(x-50) \times 255}{100} \end{aligned}$$

For $x = 50$

$$S = \frac{(50-50) \times 255}{100}$$

~~$= 0$~~
For $x = 100$

$$S = \frac{(100-50) \times 255}{100}$$

$$= \frac{50 \times 255}{100} = 127.5$$

For $7 = 150$

$$g = \frac{(150 - 50) \times 255}{100}$$

$$= \frac{100 \times 255}{100}$$

$$= 255$$

Final values

Original Intensity	Transformed Intensity
50	0
100	128
150	255

